

TOMPUDA, Russian Federation

WMO: 304390

Lat: 55.117N

Lon: 109.750E

Elev: 1525

StdP: 13.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.7	-22.2	-31.4	1.0	-26.1	-26.4	1.3	-21.5	24.9	6.5	20.7	11.4	1.3	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	74.3	60.5	70.7	59.5	67.3	59.3	63.8	68.6	62.3	67.3	60.9	65.7	3.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	89.3	65.2	60.6	83.6	64.2	58.9	78.4	62.8	30.0	69.4	28.9	67.5	27.8	65.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	14.6	10.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	-3.5	-2.6	11.0	28.5	39.6	50.1	58.4	58.7	46.8	32.1	17.4	8.8	
	DBStd	22.95	10.48	10.84	10.66	7.16	5.64	5.92	6.05	4.40	6.84	6.68	9.89	9.76	
	HDD50	8335	1658	1472	1208	646	324	68	5	2	143	555	977	1277	
	HDD65	13170	2123	1892	1673	1096	786	447	218	200	546	1020	1427	1742	
	CDD50	660	0	0	0	0	3	72	267	271	47	0	0	0	
	CDD65	19	0	0	0	0	0	0	15	4	0	0	0	0	
	CDH74	117	0	0	0	0	0	10	88	19	0	0	0	0	
	CDH80	14	0	0	0	0	0	0	13	1	0	0	0	0	
Wind	WSAvg	3.5	3.1	2.2	3.2	3.5	3.3	2.8	2.8	3.0	3.8	4.2	5.3	4.8	
Precipitation	PrecAvg	14.4	0.6	0.5	0.4	0.7	1.1	1.7	2.3	2.5	1.8	1.2	0.9	0.7	
	PrecMax	18.6	2.2	1.0	1.3	1.7	2.7	3.6	5.4	4.2	3.2	2.8	1.9	1.3	
	PrecMin	9.8	0.2	0.2	0.0	0.2	0.3	0.5	0.5	0.8	0.2	0.0	0.4	0.4	
	PrecStd	2.4	0.4	0.2	0.3	0.4	0.5	0.9	1.3	0.9	0.7	0.7	0.4	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	25.3	38.0	51.8	62.6	75.8	81.4	76.8	69.1	50.3	42.0	36.3	
		MCWB	24.8	23.0	33.6	41.4	48.4	59.8	64.2	61.9	57.2	43.6	38.4	34.1	
	2%	DB	20.0	20.8	34.2	44.8	56.7	70.2	76.2	72.1	63.4	46.3	38.5	29.2	
		MCWB	18.2	19.3	30.9	36.7	45.3	56.1	61.7	60.5	56.4	41.0	35.2	26.5	
	5%	DB	14.9	17.1	30.2	40.4	52.7	65.3	72.1	68.9	60.4	43.5	34.9	25.7	
		MCWB	13.6	15.5	27.0	34.8	42.7	54.7	60.7	60.1	54.6	38.6	31.6	23.1	
	10%	DB	10.8	13.0	26.8	37.9	48.9	60.2	68.3	66.3	57.2	41.3	31.6	22.5	
		MCWB	9.7	11.8	23.9	33.8	41.2	52.8	59.9	60.4	52.2	37.0	28.6	20.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	23.4	34.2	42.1	50.5	60.7	67.6	65.5	60.3	45.2	38.9	33.2
			MCDB	26.9	25.3	36.1	50.3	59.7	71.3	72.6	67.7	64.5	49.8	41.5	35.3
2%		WB	18.3	19.3	30.9	38.2	46.7	57.7	64.3	63.8	58.0	42.0	35.5	27.1	
		MCDB	19.8	20.8	33.8	43.0	53.9	67.5	70.2	67.2	62.3	45.4	38.2	29.1	
5%		WB	13.3	15.5	27.3	36.1	44.3	55.6	62.8	62.5	55.5	39.4	32.1	23.2	
		MCDB	14.4	16.9	30.0	39.6	50.7	63.0	69.0	66.2	59.4	42.8	34.8	25.6	
10%		WB	9.7	11.7	23.9	34.5	42.1	53.6	61.2	61.2	52.4	37.4	28.8	20.1	
		MCDB	10.6	12.7	26.6	37.7	47.5	59.6	66.8	65.1	56.9	40.5	31.5	22.5	
Mean Daily Temperature Range		5% DB	MDBR	13.8	15.5	17.2	14.3	14.1	14.6	13.8	13.6	13.5	12.1	11.6	12.1
			MCDBR	13.6	15.9	17.8	17.7	21.5	24.7	22.1	18.4	15.7	15.1	12.1	12.8
	MCWBR		12.0	14.2	14.7	11.5	12.3	13.0	10.6	9.8	9.3	11.1	11.6	12.1	
	5% WB	MCDBR	12.9	15.6	16.2	15.3	19.2	20.4	16.3	13.3	13.7	14.4	12.0	12.7	
		MCWBR	12.0	14.2	13.3	11.2	12.4	12.4	10.2	9.1	9.2	11.7	12.2	12.2	
Clear-Sky Solar Irradiance	taub		0.250	0.284	0.309	0.352	0.395	0.433	0.417	0.384	0.335	0.294	0.262	0.242	
	taud		2.081	2.065	2.087	2.141	2.172	2.223	2.323	2.392	2.462	2.352	2.107	2.035	
	Ebn at Noon		212	247	270	274	268	259	261	263	264	246	203	183	
	Edh at Noon		20	30	38	42	43	42	37	33	26	23	19	17	
All-Sky Solar Radiation	RadAvg		286	594	1071	1479	1682	1844	1677	1400	958	544	286	199	
	RadStd		23	39	52	100	124	121	165	94	76	55	29	17	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page